

- 2 (a) molecule(s) rebound from wall of vessel / hits walls B1
 change in momentum gives rise to impulse / force B1
either (many impulses) averaged to give constant force / pressure
or the molecules are in random motion B1 [3]
- (b) (i) $p = \frac{1}{3} \rho \langle c^2 \rangle$ C1
- $1.02 \times 10^5 = \frac{1}{3} \times 0.900 \times \langle c^2 \rangle$
- $\langle c^2 \rangle = 3.4 \times 10^5$ C1
 $c_{\text{RMS}} = 580 \text{ m s}^{-1}$ A1 [3]
- (ii) *either* $\langle c^2 \rangle \propto T$ *or* $\langle c^2 \rangle = 2 \times 3.4 \times 10^5$ C1
 $c_{\text{RMS}} = 830 \text{ m s}^{-1}$ (allow 820) A1 [2]
- (c) c_{RMS} depends on temperature (alone) B1
 so no effect B1 [2]

Page 3	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – May/June 2008	9702	04

2 (a) (i) amplitude = 0.5 cm

A1 [1]